

RK Health – AI Smart Patient Appointment & Medication Reminder System

Project Description:

RK Health is a web-based healthcare management application that gives patients, front-desk staff and doctors a single dashboard for appointment scheduling, medication tracking, report generation and AI-generated patient summaries. The platform includes an Appointment Management module for booking, searching, filtering and tracking appointment status; a Medication Management module for recording dosage, frequency and reminder schedules; a Reports module with CSV export and print support; and an AI Summary module that produces health insights, risk levels and medication adherence scores for each patient.

Built with HTML, CSS and JavaScript on the frontend and Google Apps Script on the backend, RK Health uses Google Sheets as its database, giving the clinic a zero-cost, easy-to-inspect data store. The application is deployed publicly via GitHub Pages, and every module — Dashboard, Appointments, Medications, Reports and AI Summary — is fully responsive across desktop and mobile.

Scenarios

Scenario 1: A front-desk staff member opens the Appointments module, searches for a patient, filters by status, and confirms three upcoming appointments — all tracked live on the Dashboard.

Scenario 2: A patient on Metformin has their dosage, frequency and reminder schedule tracked in the Medications module, with the dashboard surfacing pending reminders and active prescriptions at a glance.

Scenario 3: A doctor generates an AI Summary for a patient before a visit, instantly seeing medication adherence percentage, risk level, and a concise health summary instead of reviewing scattered records manually.

Technical Architecture:

RK Health uses a three-tier architecture. The frontend is a responsive interface built with HTML, CSS and JavaScript, covering the Dashboard, Appointments, Medications, Reports and AI Summary pages. It is hosted publicly via GitHub Pages. The frontend communicates with a Google Apps Script backend deployed as a Web App, which validates requests, performs CRUD operations, and aggregates data for the dashboard and AI Summary module. Google Apps Script reads from and writes to Google Sheets, which serves as the application's database — with dedicated sheets for Appointments, Medications and Reports.

Frontend (HTML/CSS/JS) → GitHub Pages (Hosting) → Google Apps Script (Backend) → Google Sheets (Database)

Pre-requisites:

- Google Apps Script familiarity — [Google Apps Script Documentation](#)
- HTML, CSS and JavaScript skills — [MDN / W3Schools tutorials](#)
- Google Sheets as a data store — understanding rows/columns as CRUD records
- Version control with Git — [Git Documentation](#)
- GitHub Pages for static hosting — [GitHub Pages Documentation](#)

Project Workflow:

User → Dashboard → Appointments → Medications → Reports → AI Summary → Google Sheets

Milestone 1: Project Setup (8 Story Points)

- Project planning and repository creation
- GitHub setup and folder structure creation
- Frontend layout design
- Dashboard UI development

Milestone 2: Appointment Module (13 Story Points)

- Appointment CRUD (create, read, update, delete)
- Appointment validation logic
- Search, filters and pagination
- Google Sheets integration for the Appointments sheet

Milestone 3: Medication Module (13 Story Points)

- Medication CRUD (create, read, update, delete)
- Search, filters and pagination
- Google Sheets integration for the Medications sheet

Milestone 4: Reports & AI Summary (13 Story Points)

- Reports module with searchable, filterable table
- AI Summary module for health insights and risk scoring
- Dashboard integration of live stats
- CSV export and print support
- Dynamic statistics across all modules

Milestone 5: Testing & Deployment (8 Story Points)

- CORS issue fixes
- Google Apps Script integration finalized
- Bug fixing across modules
- GitHub Pages deployment
- Final testing and documentation

Backend: Google Apps Script, CRUD & Google Sheets Integration

The Google Apps Script backend is deployed as a Web App bound to the RK Health Google Sheet. It exposes endpoints that the frontend calls using URLSearchParams-based POST requests (chosen over raw JSON POSTs specifically to avoid CORS preflight issues encountered during development).

CRUD Operations: Each module — Appointments and Medications — maps to a dedicated sheet. Create appends a new row; Read fetches and filters rows for display; Update locates a row by its unique identifier and rewrites the relevant columns; Delete removes the corresponding row. Reports are derived by reading and aggregating data from the Appointments and Medications sheets.

Google Sheets Integration: Google Sheets acts as the application's database, with each sheet (Appointments, Medications, Reports) structured as a simple table. The Apps Script backend reads and writes to these sheets directly, so every action in the frontend — booking an appointment, updating a medication schedule, generating a report — is reflected immediately in the underlying spreadsheet.

API Communication: The frontend sends requests to the deployed Apps Script Web App URL. Responses are returned as JSON and rendered into the Dashboard, tables, and AI Summary cards. Date and time formatting issues encountered between Sheets and the frontend were identified during testing and resolved.

Deployment:

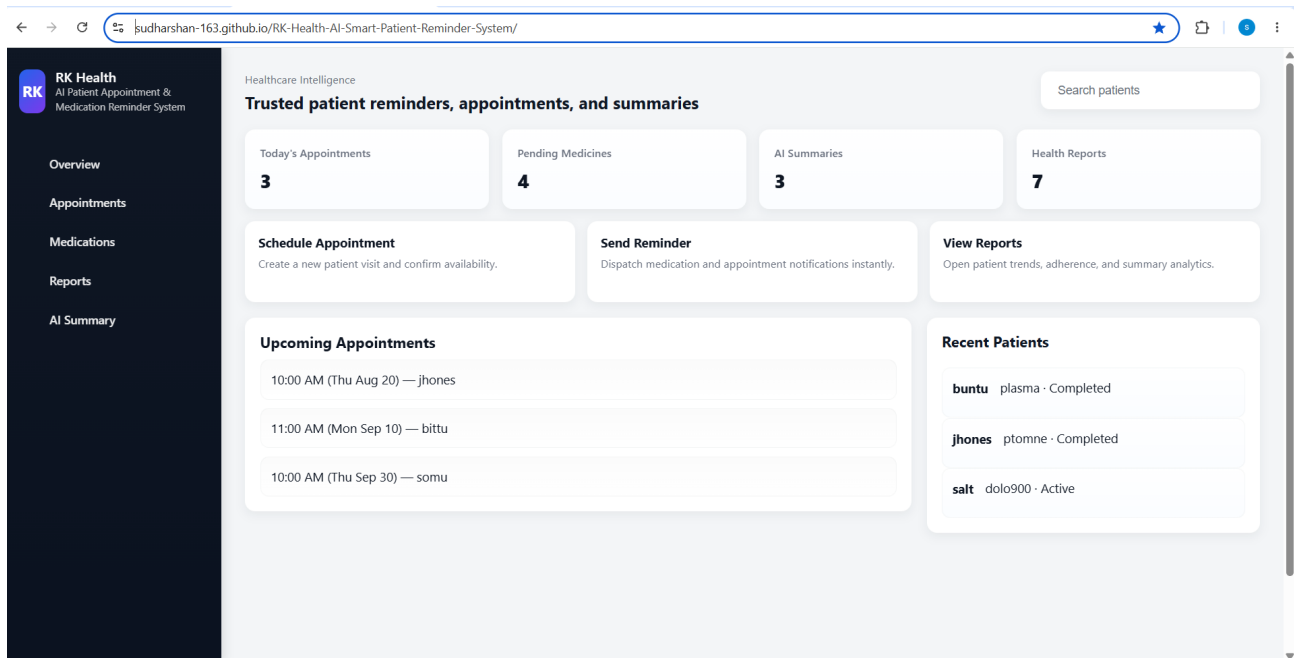
Source Code Repository: <https://github.com/sudharshan-163/RK-Health-AI-Smart-Patient-Reminder-System>

Live Deployed Application: <https://sudharshan-163.github.io/RK-Health-AI-Smart-Patient-Reminder-System/>

The frontend is deployed as a static site on GitHub Pages, directly from the project's GitHub repository. The Google Apps Script project is deployed separately as a Web App, with its deployment URL referenced by the frontend for all data operations.

Exploring RK Health's Web Pages:

Dashboard



The Dashboard is the landing page of RK Health, giving an at-a-glance view of today's appointments, pending medicines, AI summaries generated, and health reports. Quick-action cards let the user schedule an appointment, send a reminder, or jump straight into reports.

Appointment Management

The screenshot displays the 'Appointment Management' web application. The browser address bar shows the URL: `sudharshan-163.github.io/RK-Health-AI-Smart-Patient-Reminder-System/appointment.html`. The application has a dark sidebar on the left with the 'RK Health' logo and navigation links: Overview, Appointments, Medications, Reports, and AI Summary. The main content area is titled 'Appointment Management' and features a sub-header 'Keep every patient visit on schedule and under control.' Below this is a search bar and a '+ Schedule Appointment' button. A filter section includes a dropdown menu set to 'All', a date input field with the placeholder 'dd-mm-yyyy', and a 'Reset Filters' button. A summary row shows four categories: 'Today's Appointments' (0), 'Confirmed' (3), 'Pending' (0), and 'Cancelled' (0). Below the summary are three action cards: 'Schedule Appointment' (Book a new consultation and assign doctor availability instantly), 'Send Reminder' (Push appointment reminders to patients with one click), and 'Calendar View' (Review doctor schedules, appointment slots, and clinic capacity). The 'Daily Appointment Tracker' section shows 'Total records: 3' and a 'Jump to upcoming' link. A table lists the appointments with columns for Patient Name, Doctor, Department, Date, Time, Status, and Action. The table contains three rows, all with a 'Confirmed' status.

Patient Name	Doctor	Department	Date	Time	Status	Action
jhones	sham	dentist	Aug 20, 2001	10:00 AM	Confirmed	Edit Delete
bittu	ravi	cardiologist	Sep 10, 2001	11:00 AM	Confirmed	Edit Delete
somu	trump	ent	Sep 30, 2001	10:00 AM	Confirmed	Edit Delete

The Appointments page lets staff schedule new appointments and manage existing ones, with live counts for today's appointments, confirmed, pending and cancelled visits. The Daily Appointment Tracker table supports search, filtering by status/date, and inline edit/delete actions.

Medication Management

RK RK Health
AI Patient Appointment &
Medication Reminder System

Overview

Appointments

Medications

Reports

AI Summary

Medication Management

Manage prescriptions, reminders, and patient adherence in one place.

Search patient, medicine, doctor

+ Add Medication

All Statuses

Reset Filters

Today's Medicines
0

Active Prescriptions
2

Pending Reminders
2

Completed Today
0

Medication Tracker
Total records: 4

Review history

Patient	Medicine	Dosage	Frequency	Start Date	End Date	Reminder Time	Doctor	Reminder Status	Action
buntu	plasma	500	50	2027-05-09T18:30:00.000Z	2027-11-19T18:30:00.000Z	1899-12-30T16:38:50.000Z	banu	Completed	Edit Delete
jhones	ptomne	456	45	2026-11-19T18:30:00.000Z	2027-05-14T18:30:00.000Z	1899-12-30T05:38:50.000Z	ravi	Completed	Edit Delete
salt	dolo900	450	100	2026-11-19T18:30:00.000Z	2027-11-19T18:30:00.000Z	1899-12-30T00:38:50.000Z	sham	Active	Edit Delete
ichan	plasma	500	99	2027-04-	2028-06-	1899-12-	sham	Active	Edit

The Medications page tracks prescriptions, dosage, frequency, start/end dates and reminder status for each patient. Stat cards summarize today's medicines, active prescriptions and pending reminders, and the Medication Tracker table supports filtering and per-record edit/delete.

Reports

RK RK Health
AI Patient Appointment &
Medication Reminder System

Overview

Appointments

Medications

Reports

AI Summary

Reports & Analytics

View patient statistics, appointment analytics and healthcare reports.

Search reports...

+ Generate Report

All Report Types

dd-mm-yyyy

dd-mm-yyyy

Export CSV

Export PDF

Print

Total Reports

7

Patients Reviewed

6

High Risk Patients

0

Generated Today

7

Generate Health Report

Create a new comprehensive patient health report using AI summaries.

Export PDF

Download selected reports as printable PDFs.

Export Excel

Export report metadata for further analysis in spreadsheets.

Reports Table

Download All

Report ID	Patient Name	Report Type	Doctor	Date	Status	Action
APT-c9296006-963a-43b8-b0a4-6a22f8d5bd66	jhones	dentist	sham	Thu Aug 20	Confirmed	View Download Delete
APT-1783618147949	bittu	cardiologist	ravi	Mon Sep 10	Confirmed	View Download Delete

Recent Activity

AI

AI Summary

Generated blood panel summary for Emma Walker

Ge

NK

Kavita Rao

Downloaded ECG report for Noah Patel

Co

NS

Neha Sharma

Requested diabetes summary for Ava Johnson

The Reports page aggregates report data across patients, with totals for reports generated, patients reviewed and high-risk cases. Reports can be searched, filtered by date/type, exported as CSV or PDF, or printed directly, and a Recent Activity feed shows the latest report actions.

AI Summary

RK RK Health
AI Patient Appointment &
Medication Reminder System

Overview

Appointments

Medications

Reports

AI Summary

AI Patient Summary

AI-generated health insights, appointment recommendations and medication adherence analysis.

Search patient summaries...

+ Generate AI Summary

AI Summaries Generated
3

High Risk Patients
1

Medication Adherence
83%

AI Accuracy Score
92%

Generate New AI Summary

Run AI analysis across selected patient records and generate a concise summary.

Predict Health Risk

Use AI to predict near-term risks and flag patients for follow-up.

Export AI Report

Export AI summary batches to PDF or CSV for clinical review.

AI Patient Summaries

Export All

Search patient summaries

Previous Page 1 of 1 Next

EW

Emma Walker
62 · Cardiology

MEDIUM

Current Meds: Metformin 500mg Last Appointment: 2026-07-06 Next Appointment: 2026-07-20

Recent AI Activity

See all

EW

Generated Health Summary
Emma Walker · 2026-07-06

Completed

NP

Generated Risk Assessment
Noah Patel · 2026-07-05

Completed

The AI Summary page generates AI-based patient health insights, appointment recommendations and medication adherence analysis. It surfaces AI summaries generated, high-risk patients, medication adherence percentage and an AI accuracy score, alongside a searchable list of per-patient AI summaries and a live feed of recent AI activity.

Conclusion:

RK Health is a web-based healthcare management application that enables appointment scheduling, medication tracking, report generation and AI-based patient summaries, using Google Apps Script and Google Sheets as the backend. The application was deployed using GitHub Pages and demonstrates complete frontend-backend integration with CRUD operations and dynamic reporting.

Major Challenges: Google Apps Script integration, CORS errors, CRUD synchronization, Google Sheets mapping, date formatting, time formatting, frontend-backend data consistency, dynamic reports generation, AI Summary integration, GitHub Pages deployment, and extensive debugging during testing.

Team: Sudharshan Kacharla (Team Lead — project planning, frontend development, backend development, Google Apps Script, Google Sheets integration, GitHub deployment, testing, bug fixing, documentation, final integration) and Sai Raghavendra (Team Member — assisted during project discussions, participated in project review and testing).